

Simulation

Two processes:

- Bind / replace
- Bind / unbind

Simulation

```
class Simulation:

    ...

    def __simulate(self, iteration: int, timesteps: int, data: Dict) -> None:
        for timestep in range(timesteps):
            self.pre_step(iteration, timestep, data)
            self.step(iteration, timestep, data)
            self.post_step(iteration, timestep, data)
```

Bind / replace

```
# if the position on the scaffold is vacant - place tile in position
if scaffold.is_free(tile.position):
    scaffold.place_tile(tile)

# else if the position is taken, and the current tile has less or equal
# mismatches than the existing tile - replace the existing tile
elif scaffold.count_mismatches(tile) <= scaffold.count_mismatches(scaffold.get_tile(tile.position)):
    scaffold.replace_tile(tile)
```

Bind / unbind

- # 1 - chose scaffold from collection
 - # 2 - chose bound tile randomly from scaffold
 - # 3 - calculate the "Gibbs Free Energy" according to SantaLucia, Hicks
 - # 4 - what next?
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- # - if the value calculated at step 3 is < 0 then do something spontaneously? (in this context unbind?)
 - # - if the value calculated at step 3 is > 0 then do nothing?
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- # - Reaction rate?
 - # - Stochastic component?
 - # - Reaction or probability threshold?
 - # - What about other interactions?
 - # - Is GFE relevant in the context of bind / replace?